

DIKSHA PHULORIA

NJ, USA | dikshaphuloria@gmail.com | +1 (609) 286-8369 | [Portfolio](#)

SUMMARY

Data Science graduate from Rutgers University (GPA 3.9). I've always seen building solutions as solving a carefully placed puzzle and that curiosity has taken me from enterprise analytics to agentic AI systems and mechanistic interpretability research.

EDUCATION

Rutgers University

M. S. Data Science - GPA: 3.9/4

August 2024 - May 2026

New Brunswick, NJ, USA

Manipal University Jaipur

B. Tech, Information Technology - GPA: 8.07/10

July 2017 - June 2021

Rajasthan, India

WORK EXPERIENCE

Accenture

Analyst, Custom Software Engineering

May 2021 - June 2023

Gurugram, India

- Supported enterprise analytics platforms like Tableau and Cognos, which improved reporting reliability and performance.
- Analyzed and resolved over 100 client-reported issues using ServiceNow by debugging data pipelines and application workflows.
- Monitored issue lifecycle metrics to ensure compliance with service level agreements and improve client satisfaction.

PROJECTS

NutriSync - Agentic AI Meal Planning System

February 2026 - May 2026

Next.js 15, FastAPI, PostgreSQL, Google Gemini, Docker, TypeScript, Tailwind CSS

- Architected a full-stack agentic AI platform unifying meal planning, nutrition tracking, pantry management, and grocery shopping with Google Gemini, using Human-in-the-Loop (HITL) checkpoints at every decision stage for reliability in a health-critical context.
- Engineered seamless cross-module data propagation: selecting a meal auto-deducts pantry inventory, recalculates macros, and refreshes the grocery list all without additional user input.
- Integrated Gemini Vision for AI receipt scanning; backed by a 12-table PostgreSQL schema seeded from Spoonacular and USDA FoodData Central APIs, with a freemium model supporting both guest and authenticated experiences.

Interpreting Climate Representations in GPT-2 via Sparse Autoencoders

February 2026 - May 2026

Python, PyTorch, Hugging Face Transformers, Google Colab (A100 GPU)

- Applied Sparse Autoencoder (SAE) analysis to GPT-2 Medium (354M parameters) to investigate how climate concepts are internally encoded, extending Anthropic's mechanistic interpretability framework to a specialized domain.
- Collected 4,096-dimensional MLP activations from layer 17 across 839,773 tokens from the ClimateEval dataset (30,000 samples), training an SAE with 32,768 features (8× expansion) to stable convergence at ~230 active features per token.
- Discovered and categorized 21 climate-related features across four selectivity tiers; top Climate-Selective features included Carbon (15.96× selectivity ratio) and Greenhouse/Emissions (8.56×).
- Revealed that intuitively climate-adjacent terms (Glaciers, Species, Atmosphere) are non-climate-selective in GPT-2, demonstrating that surface-level token inspection is insufficient and selectivity analysis against labeled data is essential.

Veralyze - AI Video & Image Authenticity Detector

April 2026

FastAPI, Next.js, TypeScript, Google Gemini Vision, FFmpeg, NumPy | Personal Project

- Built a forensic web application detecting AI-generated or manipulated content in videos, images, and YouTube URLs using a four-signal pipeline: Gemini Vision per-frame analysis, consecutive-frame transition detection, and Error Level Analysis (ELA).
- Designed a clean architectural split between raw signal collection (analyzer.py) and all verdict/confidence scoring logic, with duration-aware FFmpeg frame sampling that dynamically scales coverage versus API cost.
- Delivered real-time analysis feedback via Server-Sent Events (SSE) across a 15–40 second pipeline, surfacing forensic evidence proof frames, pixel scores, artifact lists so users understand exactly why content was flagged.

SKILLS

Languages and Querying: Python, SQL, R

Machine Learning and AI: Predictive Modeling, NLP, AI and LLMs, Causal Inference, Feature Engineering

Data Analysis and Visualization: Pandas, Exploratory Data Analysis (EDA), Data Cleaning, Data Visualization, ETL, Data Wrangling

Databases and Platforms: PostgreSQL, MySQL, Snowflake, Streamlit, Google Colab, ServiceNow, GIT, VS Code

CERTIFICATIONS

Artificial Intelligence Fundamentals - IBM

May 2026